

Local Vision Model Benchmark

Sales-slip OCR · 2026-06-08 12:18 · NVIDIA RTX A2000 8GB Laptop GPU, 8192 MiB

Task: extract the total amount from German grocery/gas sales slips. Each model receives the raw JPEG (resized to ≤ 1500 px) and a fixed prompt. Accuracy = fraction of the 3 test images where the extracted value matches the ground truth encoded in the filename (e.g. *slip0_7949.jpg* $\rightarrow$ 79,49 €).

Summary

Model	VRAM	Pull (s)	Accuracy	Avg latency (s)	Status
Moondream 2	~1.7 GB	1	67% (2/3)	0.3	ok
LLaVA-Phi3	~2.9 GB	1	33% (1/3)	1.8	ok
Gemma 3 4B	~3.3 GB	cached	33% (1/3)	3.2	ok
LLaVA 1.5 7B	~4.7 GB	cached	0% (0/3)	1.5	ok
German-OCR-3	~2.7 GB	23	33% (1/3)	73.4	ok
SmoVLM2 2.2B	~2.0 GB	45	0% (0/3)	0.6	ok
MiniCPM-o 4B	~5.0 GB	-	-	-	skipped: pull failed: pull model manifest: file d
Qwen3-VL 4B	~3.4 GB	112	100% (3/3)	15.3	ok
Qwen2.5-VL 7B	~5.5 GB	-	-	-	skipped: pull failed: pull model manifest: file d
Qwen2-VL 7B	~5.5 GB	-	-	-	skipped: pull failed: pull model manifest: file d
MiniCPM-V 2.6 8B	~6.0 GB	188	67% (2/3)	4.4	ok
BakLLaVA 7B	~7.4 GB	163	33% (1/3)	1.5	ok
Llama 3.2-Vision 11B	~7.9 GB	-	-	-	skipped: pull failed: write /usr/share/ollama/o

Per-model responses

Moondream 2 (moondream)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	€79.49	79,49	✓	0.3s
slip1_2841.jpg	28,41	€28.41	28,41	✓	0.3s
slip2_1093.jpg	10,93	€79.49	79,49	✗	0.3s

LLaVA-Phi3 (llava-phi3)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	85,012	85,01	✗	4.0s
slip1_2841.jpg	28,41	E28,41	28,41	✓	0.6s
slip2_1093.jpg	10,93	11,03	11,03	✗	0.6s

Gemma 3 4B (gemma3:4b)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	12,69	12,69	✗	5.6s
slip1_2841.jpg	28,41	Euro,95	NaN	✗	2.1s
slip2_1093.jpg	10,93	10,93	10,93	✓	2.1s

LLaVA 1.5 7B (llava:7b)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	NaN	NaN	✗	2.7s
slip1_2841.jpg	28,41	NaN	NaN	✗	0.9s
slip2_1093.jpg	10,93	NaN	NaN	✗	0.9s

German-OCR-3 (Keyvan/german-ocr-3)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49		NaN	✗	93.8s
slip1_2841.jpg	28,41	28,41	28,41	✓	27.7s
slip2_1093.jpg	10,93		NaN	✗	98.7s

SmolVLM2 2.2B (richardyoung/smolvlm2-2.2b-instruct)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	[img-0]The sum to pay is 123.45 Euro,Cent.	123,45	✗	1.4s
slip1_2841.jpg	28,41	[img-0]The sum to pay is 79.49 Euros.	79,49	✗	0.2s
slip2_1093.jpg	10,93	[img-0]The sum to pay is 79.49 Euros.	79,49	✗	0.2s

MiniCPM-o 4B (minicpm-o:4b)

Skipped — pull failed: pull model manifest: file does not exist (status code: -1)

Qwen3-VL 4B (qwen3-vl:4b)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	79,49	79,49	✓	29.1s
slip1_2841.jpg	28,41	28,41	28,41	✓	5.7s
slip2_1093.jpg	10,93	10,93	10,93	✓	11.0s

Qwen2.5-VL 7B (qwen2.5-vl:7b)

Skipped — pull failed: pull model manifest: file does not exist (status code: -1)

Qwen2-VL 7B (qwen2-vl:7b)

Skipped — pull failed: pull model manifest: file does not exist (status code: -1)

MiniCPM-V 2.6 8B (minicpm-v)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	66,80	66,80	✗	6.0s
slip1_2841.jpg	28,41	28,41	28,41	✓	3.5s
slip2_1093.jpg	10,93	10,93	10,93	✓	3.6s

BakLLaVA 7B (bakllava)

Image	Expected	Response (truncated)	Parsed	Match	Latency
slip0_7949.jpg	79,49	79,49	79,49	✓	2.6s
slip1_2841.jpg	28,41	79,49	79,49	✗	0.9s
slip2_1093.jpg	10,93	79,49	79,49	✗	0.9s

Llama 3.2-Vision 11B (llama3.2-vision:11b)

Skipped — pull failed: write

/usr/share/ollama/.ollama/models/blobs/sha256-9999d473417a8e179d993498195be5f42cab963acc75f4a6b15d981e8b68abed-partial: no space left on device (status code: -1)

Prompt used: "What is the sum to pay in the given sales slip? It is a German sales slip for groceries or gas. Look for 'Summe', 'Gesamt' or 'zu zahlen'. Reply with ONLY the amount in the format 'Euro,Cent' (e.g. '79,49'). No currency symbol, no extra text. If not found, reply 'NaN'."